

Human-in-the-Loop

GLOSSARY 02 - SHEET 6 / 8

A control design in which a human decision is required within the system's operating cycle

PROFESSIONAL DEFINITION

Human-in-the-loop is a control design in which a human being is positioned inside the operating cycle of an automated system so that the system cannot complete a defined action without human input, review or approval. It is distinguished from human-on-the-loop, where the human supervises and may intervene but the system proceeds by default, and from human-in-command, where humans retain authority over whether the system operates at all (Amershi et al., 2014; HLEG AI, 2019).

WHO DEFINED IT ORIGINALLY

The design pattern originates in mid-twentieth-century control engineering and human-factors research, and was carried into machine learning through interactive machine learning, formalised by Amershi, Cakmak, Knox and Kulesza (2014). The three-way distinction used in European governance was set out by the High-Level Expert Group on AI in the Ethics Guidelines for Trustworthy AI (2019) and underpins the human oversight requirement in Article 14 of the AI Act.

WHY THIS DEFINITION WAS DEVELOPED

Early automation debates treated oversight as binary - automated or not - which gave designers no vocabulary for the intermediate arrangements that most real systems use. Naming three distinct positions made it possible to specify oversight at the level of the action rather than the system, and to require different arrangements for different risk levels.

FIGURE 1 - THREE POSITIONS OF HUMAN OVERSIGHT

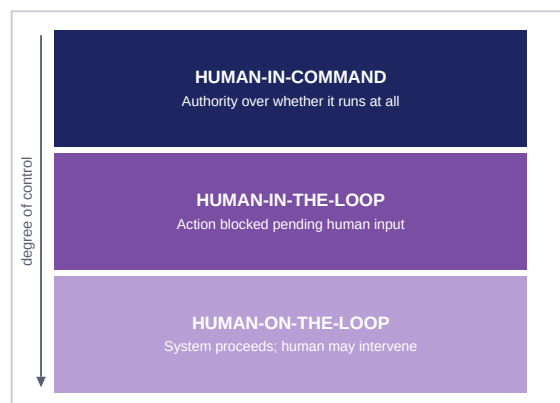


Figure 1. The three positions differ in what happens when the human does nothing. Only in the middle position does inaction stop the action, which is why it is the arrangement required for consequential decisions (after HLEG AI, 2019).

HOW IT IS USED

Specify, for each consequential action, which of the three positions applies, and verify that the human in that position has the information, the time and the authority to dissent; oversight without any of the three is nominal.

WHEN IT IS USED

At the design of any high-risk AI system, in conformity assessment under Article 14 of the AI Act, after an incident in which a human approved an output without examining it, and when automation bias is suspected.

BY WHOM IT IS USED

System designers and product owners specifying controls, risk and compliance functions evidencing oversight, operators occupying the loop, and auditors testing whether the control is real.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor concept was the manual override, a mechanism available in principle and unused in practice because the operator lacked the information or the time to exercise it. The in-the-loop, on-the-loop and in-command distinction relocates the design question from whether a human can intervene to whether the process requires them to, and pairs each position with an evidential burden. Article 14 then made that burden legally testable rather than a matter of design intent.

RELATED DEFINITIONS IN THIS COURSE

AI Literacy - Unit 2

Algorithm Aversion - Unit 2

Explainability - Unit 6

AI Governance - Unit 6

Accountability Gap - Unit 6

SOURCES (HARVARD REFERENCING STYLE)

Amershi, S., Cakmak, M., Knox, W.B. and Kulesza, T. (2014) 'Power to the people: the role of humans in interactive machine learning', AI Magazine, 35(4), pp. 105-120.

European Parliament and Council (2024) Regulation (EU) 2024/1689, Article 14 (human oversight). Official Journal of the European Union, L, 12 July 2024.

High-Level Expert Group on Artificial Intelligence (2019) Ethics Guidelines for Trustworthy AI. Brussels: European Commission.

Parasuraman, R. and Riley, V. (1997) 'Humans and automation: use, misuse, disuse, abuse', Human Factors, 39(2), pp. 230-253.